

Module: V

Electrical Installations

- Components of LT Switchgear: Switch Fuse Unit (SFU), MCB, ELCB, MCCB
- Types of Wires and Cables
- Earthing.
- Types of Batteries, Important Characteristics for Batteries.

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COMPONENTS OF LT (LOW TENSION) SWITCHGEAR

The apparatus used for switching, controlling and protecting the electrical circuits and equipment is known as switchgear. The term 'switchgear' is a generic term encompassing a wide range of products like circuit breakers, switches, switch fuse units, off-load isolators, HRC fuses, contactors, earth leakage circuit breakers, etc.

- **Switch Fuse Unit (SFU)**
- **Miniature Circuit Breaker (MCB)**
- **Moulded Case Circuit Breaker (MCCB)**
- **Earth Leakage Circuit Breaker (ELCB)**

Switch Fuse Unit (SFU)

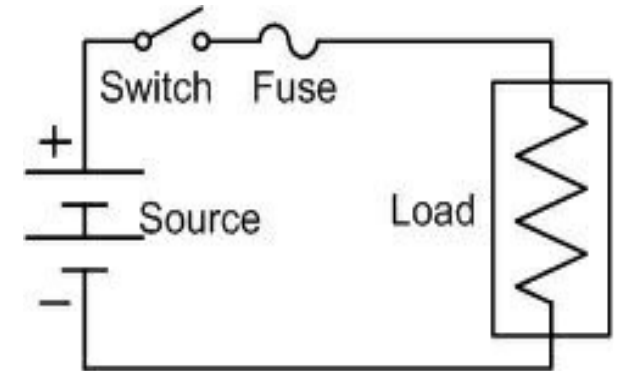
- Combination of switch and fuse together is called SFU.
- A switch isolate the circuit from supply for repair and maintenance. It is manually operated.
- Fuse is a protective device which acts quickly during abnormal condition to disconnect the circuit from the supply.

The working principle of the fuse is “ heating consequence of the current”. It is fabricated with a lean strip for thread of metallic wire. The connection of the fuse in an electrical circuit is always in series.

When the too much current is produced due to the heavy flow of current in the electrical circuit the fuse get soft and it opens the circuit. The extreme flow of current main direct to the collapse of the wires and prevents the supply.

Diffuse can be changed by the new fuse with an appropriate rating. It can be designed with elements like copper, zinc, aluminium and silver.

They also perform like a circuit breaker for breaking the circuit while the abrupt fault happens in the circuit. This works like a safety measure for protector for humans from risk



SWITCH FUSE UNIT (SFU)

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Good Fuse



Blown Fuse



Switch fuse unit



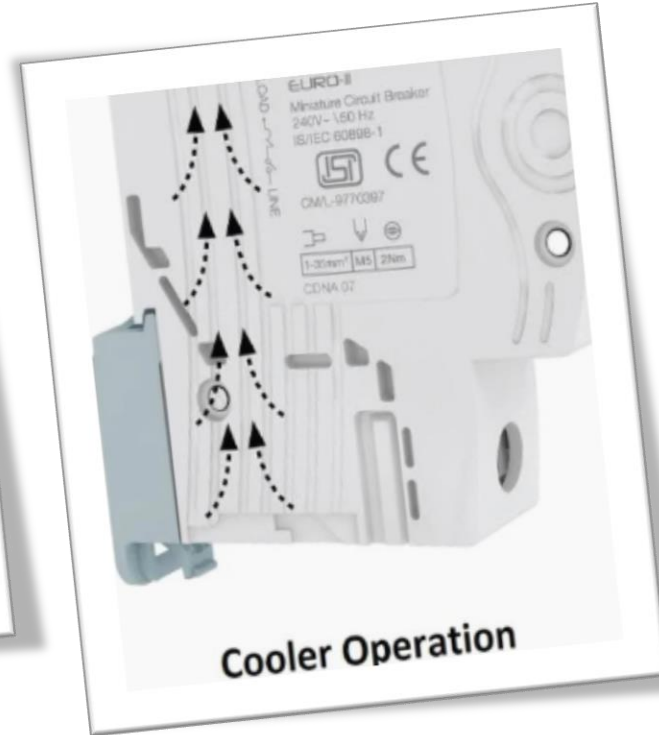
MINIATURE CIRCUIT BREAKER (MCB)

- An electromechanical device which makes and breaks the circuit in the normal operation.
- During fault/abnormal condition when current exceeds the preset value, MCB disconnects the circuit.
- MCB is high fault capacity current limiting, trip free, automatic switching with thermal and magnetic operation.
- MCB provides protection against overload and short circuit.
- MCBs are rated for 240V single phase, 415V three phase, and 220 V dc. The current rating is 0.5A to 63A.

MCBs are used because of following features:

- Its operation is very fast.
- No tripping circuit is required
- Automatic operation.
- Protection against overload and short circuit without noise, flames or smoke.
- Reset quickly after fault correction.
- No rewiring is required.
- Cannot be reclosed if fault exist.

MINIATURE CIRCUIT BREAKER (MCB)



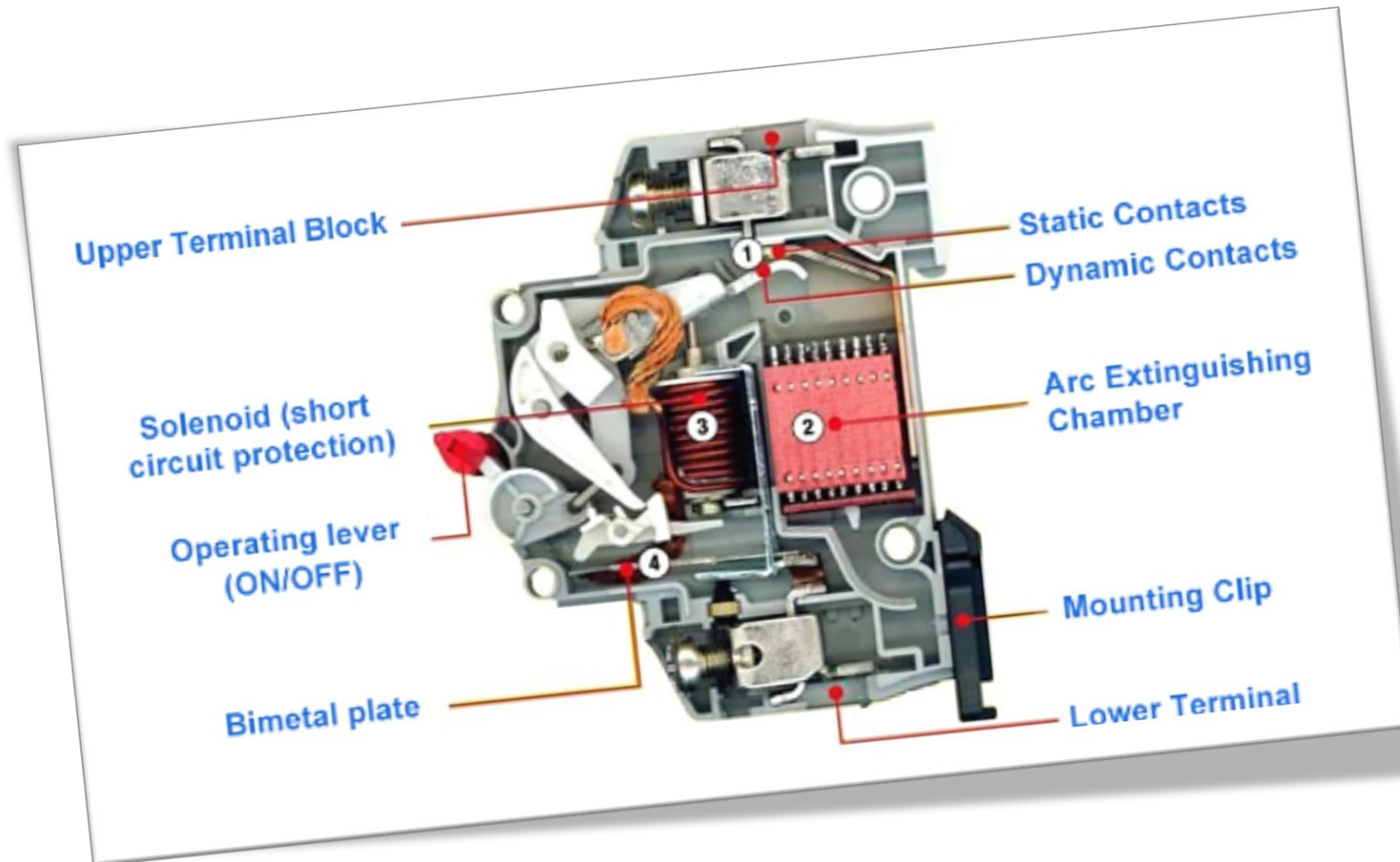
3-POLE MCB (Different Views)

MINIATURE CIRCUIT BREAKER (MCB)

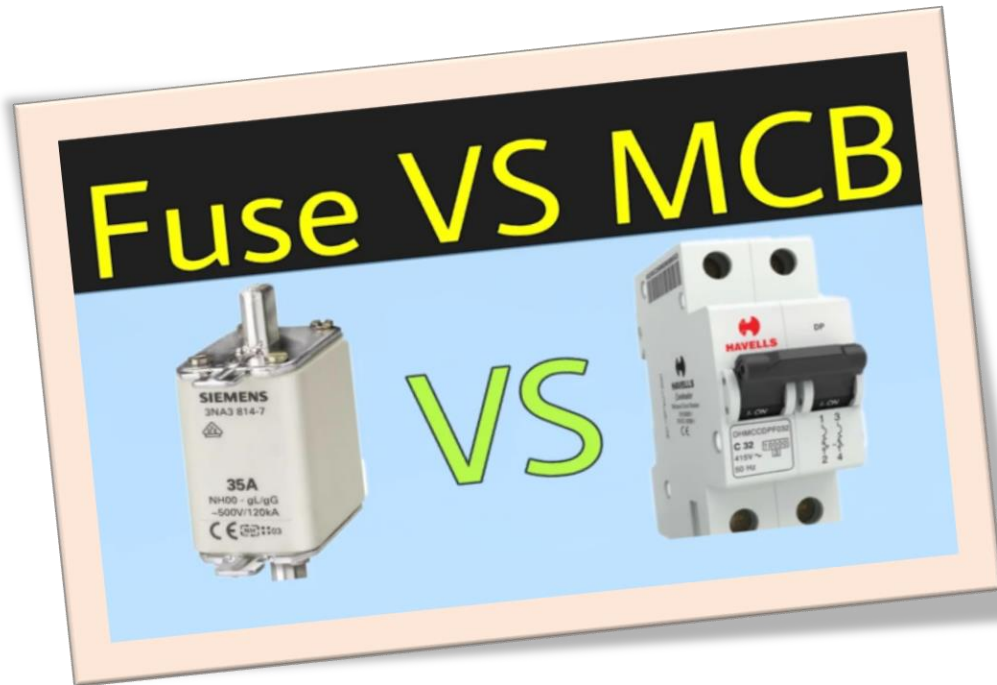


2-POLE MCB)

MINIATURE CIRCUIT BREAKER (MCB)



Comparison between Fuse and MCB



Category	MCB	Fuse
Definition	A circuit protection device that trips when there is excess current flow	Protective device that self-destructs when the current surpasses a preset value in the circuit
Working mechanism	Dual mechanism based on both electromagnetic and thermal principles	Operates by utilizing the thermal properties of conducting materials
Speed	Slower	Faster
Tripping element	Electromagnetic core with a plunger arrangement	Metallic wires
Reusability	Usually, can be reused	Usually, cannot be reused
Voltage Suitability	Low voltage circuit protection	Suitable for varying voltages from very low to very high
Applications	Lighting circuits, residential setups, and some industrial systems	Wide range of applications from small electronic circuits to heavy-duty electric motors
Cost	More expensive	Less expensive

MOULDED CASE CIRCUIT BREAKER (MCCB)

- MCCB is similar to MCB but used for circuit having current range from 63A to 3000A.
- Working is based on thermal mechanism. The bimetallic contact expands and contract when there are temperature changes.
- Under normal condition, the contacts are closed allowing current to pass. Under over-load or short circuit condition, current exceeds its safe value. Due to this, heat is generated, and the contacts are opened to interrupt the circuit.
- Due to the interruption of high current, there is arc formation. Hence in MCCB there are arc extinguishers which suppress the arc.



MOULDED CASE CIRCUIT BREAKER (MCCB)

How it works

- MCCBs detect overloads or short circuits and automatically cut off the power supply to prevent damage to the electrical system.

Components

- MCCBs have several components, including a thermal element for overload protection, a magnetic component for short-circuit detection, and a trip unit to break the circuit.

Applications

- MCCBs are used in a variety of settings, including industrial, commercial, and residential. They can be used in both low-voltage and high-voltage systems.

Features

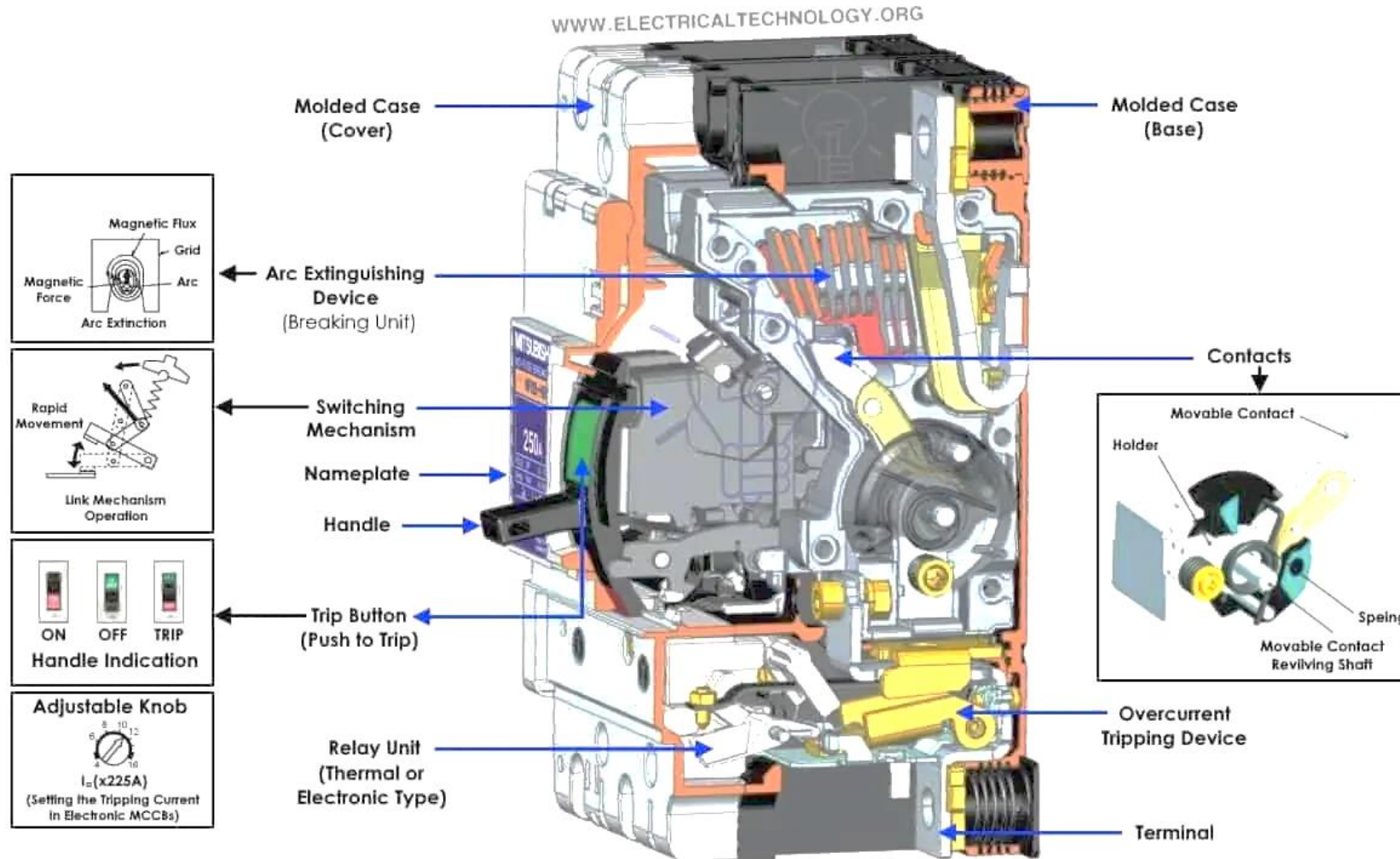
- MCCBs offer adjustable trip settings, thermal protection, and high interrupting capacity.

Installation

- MCCBs are typically installed in the distribution panel and provide a visible indication to let users know when the power is off.
- When visually inspecting an MCCB, you should look out for deformed contacts or cracks in the casing or insulation. You should also treat any burn marks on the contact or casing with caution.

MOULDED CASE CIRCUIT BREAKER (MCCB)

Construction of Molded Case Circuit Breaker (MCCB)



EARTH LEAKAGE CIRCUIT BREAKER (ELCB)

An earth-leakage circuit breaker (ELCB) is a safety device used in electrical installations with high Earth impedance to prevent shock. It detects small stray voltages on the metal enclosures of electrical equipment, and interrupts the circuit if a dangerous voltage is detected.

Once widely used, more recent installations instead use residual-current devices (RCDs, RCCBs or GFCIs) which instead detect leakage current directly.

The ELCB is used to protect the circuit from the electrical leakage.

When someone gets an electric shock, then this circuit breaker cuts off the power for protecting the personal safety and the circuit against short circuit and overload.

ELCB is a security device used in electrical system with high Earth impedance to avoid shock. It notices difference in current between line and neutral wire, and interrupt the circuit if an unsafe current is detected.

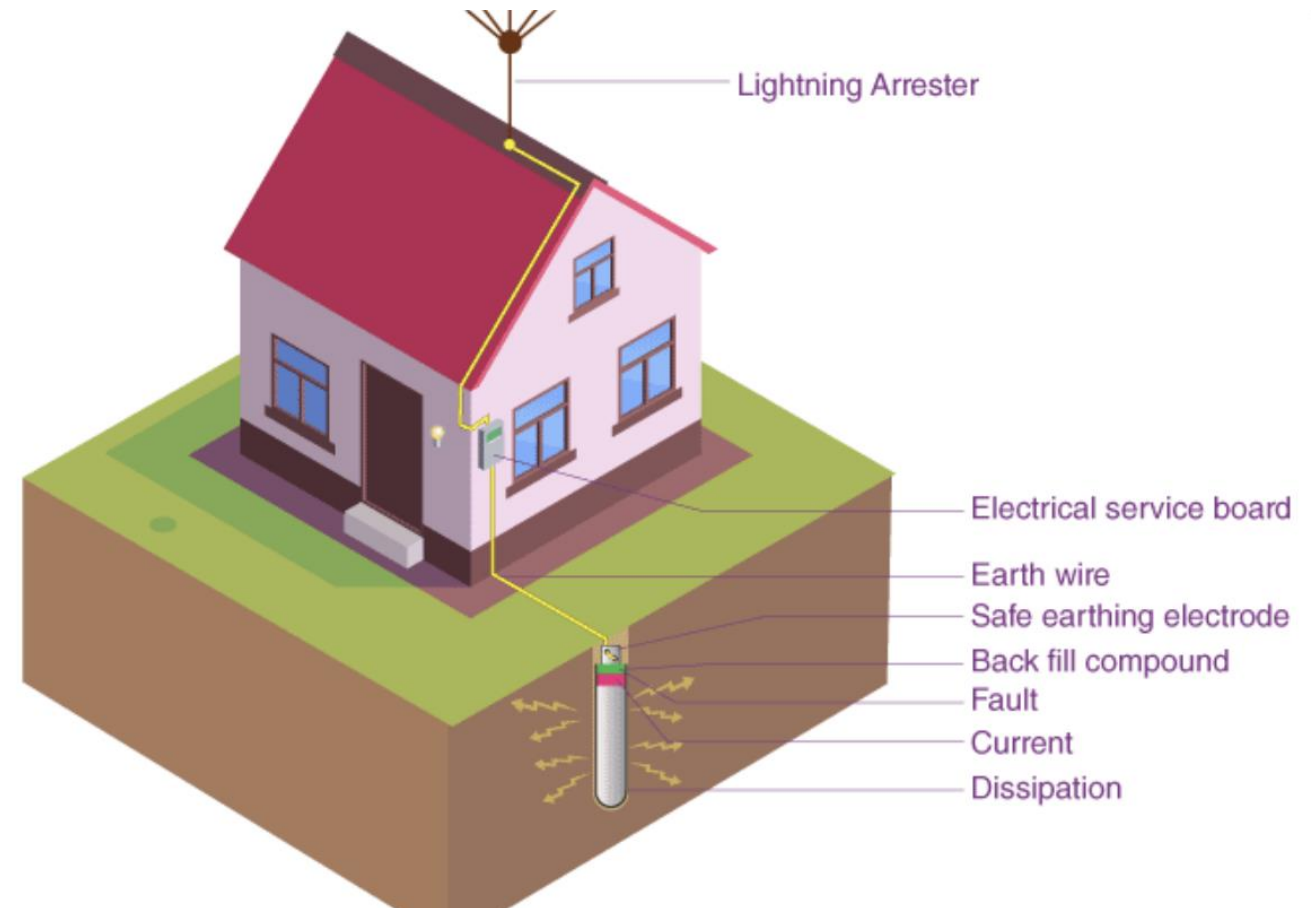


EARTHING

Earthing is defined as “the process in which the instantaneous discharge of the electrical energy takes place by transferring charges directly to the earth through low resistance wire.”

To ensure safety, earthing can be done by connecting the electrical appliance to earthing systems or electrodes placed near the soil or below the ground level.

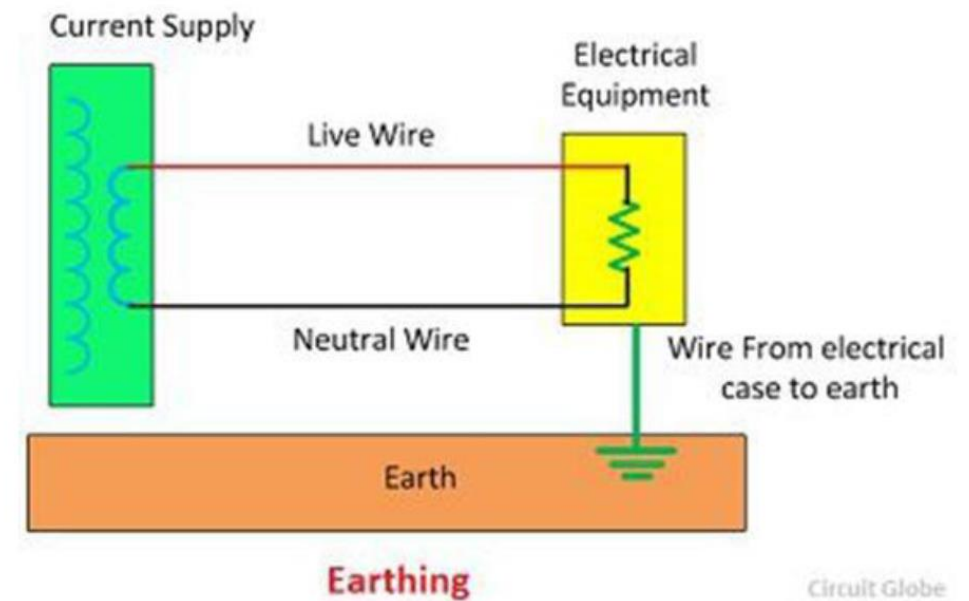
The electrode or earthing mat equipped with a flat iron riser is installed under the ground level. It helps to connect all the non-current-carrying metallic parts of the equipment.



EARTHING

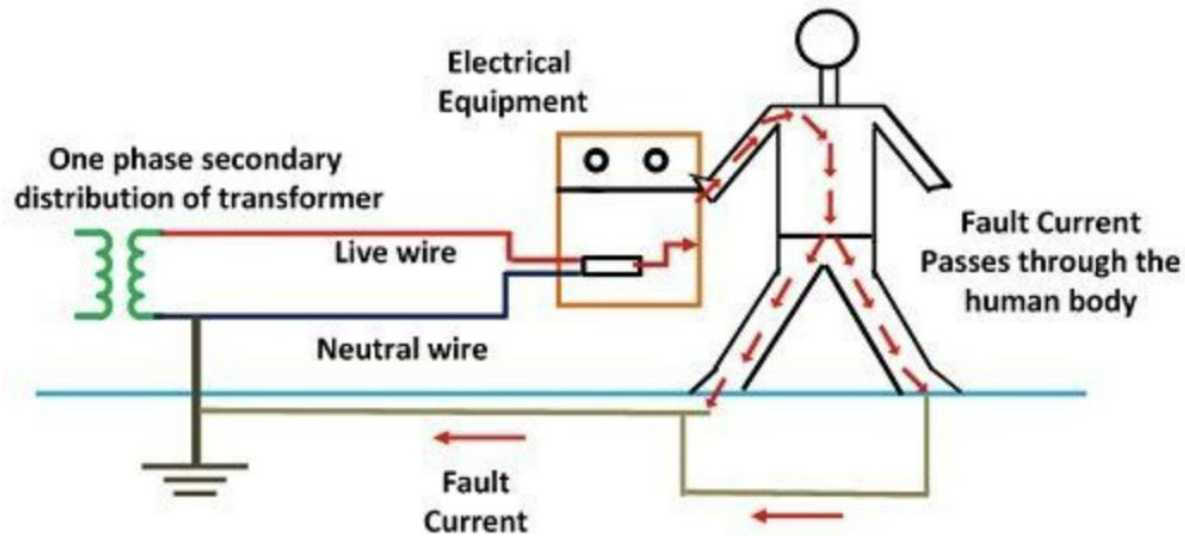
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- The connection of electrical machinery to the general mass of earth, with a conducting material of very low resistance is called earthing or grounding.
- The earthing of electrical equipment brings the equipments to zero potential and avoid the shock to the operator, under nay fault conditions.



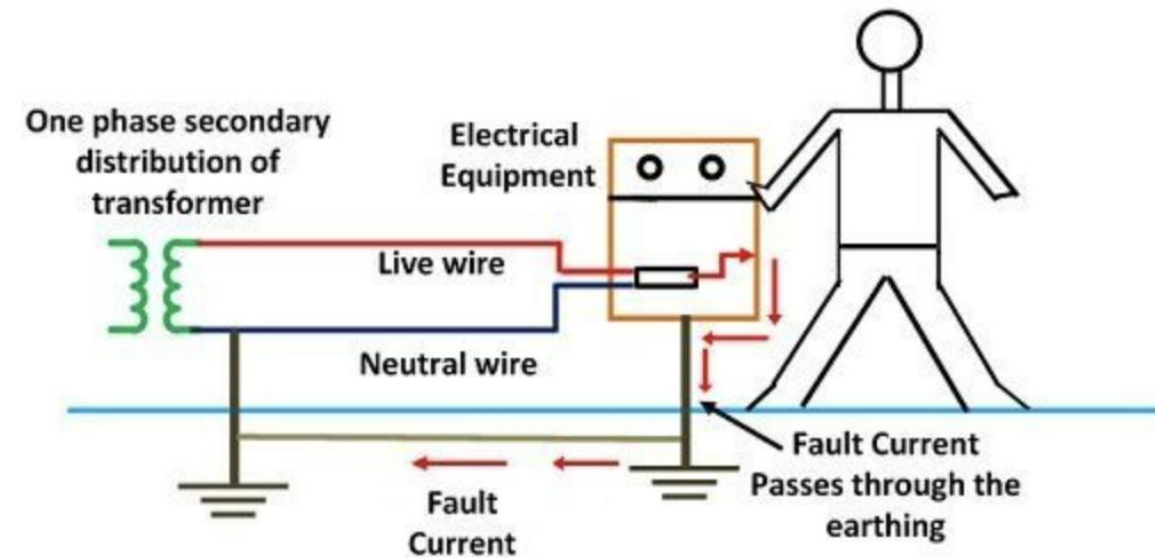
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Electrical System Without Earthing

Circuit Globe



Electrical System With Earthing

Circuit Globe

EARTHING

Earthing is defined as “the process in which the instantaneous discharge of the electrical energy takes place by transferring charges directly to the earth through low resistance wire.”

There are several types of earthing, including:

- ❖ Pipe earthing

A galvanized steel pipe is buried vertically into the ground to connect electrical conductors to the earth. This method is affordable and low maintenance, and is suitable for areas with high soil resistivity.

- ❖ Plate earthing

A metal plate, usually made of copper or galvanized iron, is buried vertically into the ground. This method is commonly used in residential and small commercial buildings.

- ❖ Strip earthing

Also known as wire earthing, this method involves burying strip electrodes in horizontal trenches. This method is often preferred for rocky terrains.

- ❖ Rod earthing

A metal rod, usually made of copper or galvanized iron, is driven vertically into the ground. This method is effective in areas with low soil resistivity, and is suitable for various environments.

PIPE EARTHING

The pipe earthing is better than another type of earthing, because it can earth more leakage current, and the possibility of braking in the earth wire is minimum.

Pipe Earthing:

- In this method the galvanized steel and perforated pipe of approved length and diameter is placed upright in a permanently wet soil
- The size of the pipe depends upon the current to be carried and type of soil
- The size of the pipe used for earthing is of diameter 40 mm and 2.5 meters in length for ordinary soil or of greater length in case of dry and rocky soil
- The depth at which the pipe must be buried depends on the moisture of the ground
- The pipe is placed at 3.75 meters
- The bottom of the pipe is surrounded by small pieces of coke or charcoal at a distance of about 15 cm
- Alternate layers of coke and salt are used to increase the effective area of the earth and to decrease the earth resistance respectively
- Generally, 10 kg of charcoal and 10 kg of salt are used for this purpose
- Another pipe of 19 mm diameter and a minimum length of 1.25 meters are connected at the top of GI pipe by reducing the socket

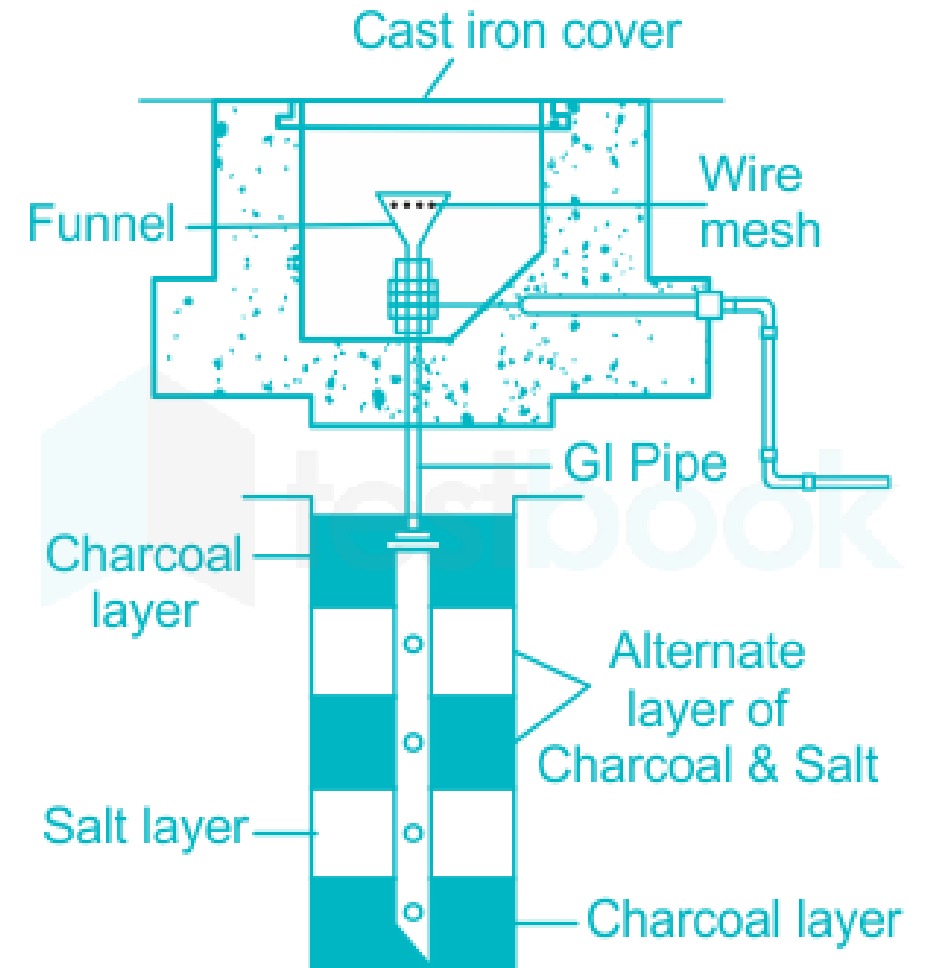
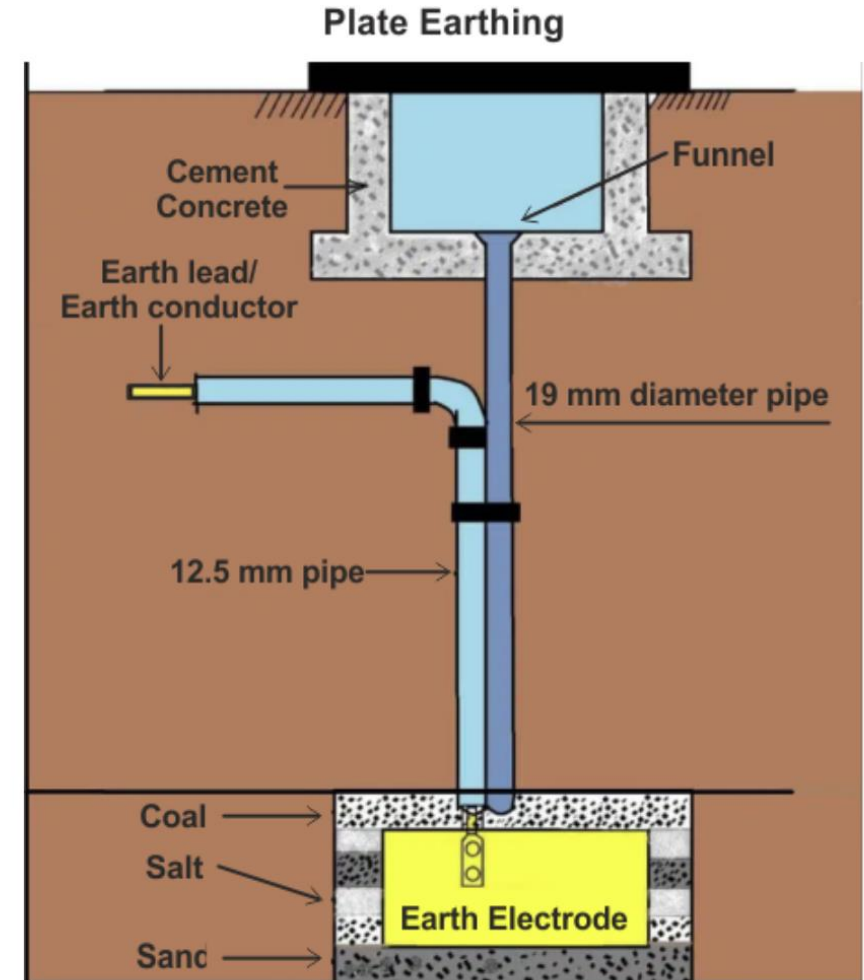


PLATE EARTHING

- The earth connection is provided with the help of **copper plate or Galvanized Iron (G.I.) plate**. The copper plate size is $60\text{ cm} \times 60\text{ cm} \times 3.18\text{ mm}$, while G.I. plate size is not less than $60\text{ cm} \times 60\text{ cm} \times 6.3\text{ mm}$. The plate is embedded 3 m (10 ft.) into the ground. The plate is kept with its face vertical.
- The plate is surrounded by **the alternate layer of coke and salt for minimum thickness** of about 15 cm. The earth wire is drawn through G.I. pipe and is perfectly bolted to the earth plate. The nut and bolts must be copper plate and must be of galvanized iron for G.I. plate.
- The earth lead used must be G.I. wire or strip of sufficient cross sectional area to carry the fault current safely. The earth wire is drawn through G.I. pipe of 19 mm diameter, at about 60 cm below the ground.
- The G.I. pipe is fitted with a funnel on the top. In order to have an effective earthing, **salt water is poured periodically through the funnel**.
- The earthing efficiency, increases with the increases of the plate area and depth of embedding. If the resistivity of the soil is high, then it is necessary to embed the plate vertically at a greater depth into ground.
- **The only disadvantage of this method is that the discontinuity of the earth wire from the earthing plate below the earth can not be observed physically. This may cause misleading and may result into heavy losses under fault condition.**



Thank you